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When Are Health Inequalities Unjust?

The Social Determinants of Health

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When is an inequality in health status between different socioeconomic groups unjust? More generally, when is an inequality in health status between different demographic groups unjust? An account of just health should help us determine which health inequalities are unjust and which are acceptable. Recognizing these distinctions will help us to remedy injustice and establish just institutions.

These versions of the second Focal Question (see Chapter 1) involve issues that move beyond simply knowing why health is of special moral importance and how that importance can guide us in extending general theories of justice to matters of health and its distribution. Knowing that protecting health protects opportunity does not, by itelf, tell us when inequalities in health, like inequalities in opportunity more generally, are unjust. Health is produced not just by having access to medical prevention and treatment but also, to a measurably great extent, by the cumulative experience of social conditions across the lifecourse. When a sixty-year old patient presents to the emergency room with a heart attack to receive medical treatment, that encounter represents the results of bodily insults that accumulated over a lifetime. Medical care is, figuratively speaking, "the ambulance waiting at the bottom of the cliff."

We have known for over 150 years that an individual's chances of life and death are patterned by social class: The more affluent and better educated people are, the longer and healthier their lives (Villerme 1840, cited in Lynch et al. 1998). When these inequalities are present in poor, developing countries, it is tempting to think that poverty and deprivation adequately explain them. We should avoid the temptation. Some rather poor countries pursue policies that produce excellent aggregate health outcomes while

In this chapter I draw heavily on Daniels, Kennedy, and Kawachi (1999, 2000). I appreciate the willingness of my coauthors to let me use that material in this chapter, and I also thank Ichiro Kawachi for the discussion of new material in it.

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moderating significantly the degree of health inequality. In addition, these effects of socioeconomic status (SES) are present as well in rich, developed countries, where the sources of inequality are not deprivation and deep poverty. More striking, these patterns persist when there is universal access to health care – a fact quite surprising to those who think that financial access to medical services is the primary determinant of health status.

In addition to socioeconomic inequalities[1] that are correlated with health inequalities, we have extensive evidence about patterns of racial and ethnic disparities in many countries. Many of these inequalities are due to the way in which social exclusion and patterns of discrimination push ethnic minorities (or majorities, as in some former colonies, like South Africa) into the lowest socioeconomic groups. Still, there is often a significant residual effect of race, as in the United States, even when correlations are controlled for income, education, and insurance levels. There are also striking patterns of gender inequality in health in some countries – leading Sen (1990) to ask where the millions of "missing" women are in some Asian countries, as compared to gender ratios in developed countries. These forms of health inequalities are the focus of the broader version of my second Focal Question: When are inequalities in health status between different groups unjust?

Of course, we cannot infer *causation* from *correlation*, and correlations alone will not support judgments about injustice. To know whether one inequality is a cause or determinant of another, and then to ascribe some judgment about inequity to it, we have to know something about the mechanisms at work. We need to know, for example, that although poor health or disability can lead to lower income, which can be a significant consequence in many settings, such "health selection," as it is called, explains only part of the correlation between low income and poor health in general,[2] and we have good reason to suppose the existence of mechanisms that work in the other direction, from socioeconomic inequality to health differences. Being less well educated, for example, can lead to engaging in some risky patterns of behavior, and these risks increase health inequalities.[3] But even

[1] Deaton (personal communication, July 13, 2006) cautions against thinking that the various components of SES (income, education, social class) all work the same way and that the convenience of combining them may be outweighed by the inaccuracy that can result. See also Deaton (2002a,b).

[2] Deaton (personal communication, July 13, 2006) and Case and Deaton (2003) point to the impact of ill health on early retirement and thus on income reduction; Case and Deaton argue that the significantly worse self-reported health of those in the lowest income quartile compared to those in the highest is primarily driven by the ill health of those out of work, presumably because they are ill. Health selection may have a greater effect on income than class. Cf. Marmot (1994).

[3] Behavior risk factors nevertheless contribute significantly to inequality: Jha et al. (2006) show that smoking contributes nearly half of the male mortality in the lowest social class in four countries.

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this mechanism fails to explain the general effect of socioeconomic inequality on health (Marmot 2004), and researchers are exploring more promising mechanisms. But even if more work is needed to clarify the exact mechanisms, it is not unreasonable to talk here about the social "determinants" of health (Marmot 1999). To the extent that these social determinants are socially controllable, we clearly face questions of distributive justice.

In some ways, gender and race or ethnic inequalities are easier to address than class inequalities. Since these kinds of inequalities are the result of social exclusion and other unjust practices aimed at vulnerable groups, we are generally and justifiably inclined to view them as inequities. Identifying them, however, does not make them easy to remedy.

Socioeconomic inequalities pose harder questions since few people are radical egalitarians opposing all forms of such inequalitiy. Many who are not at all troubled by significant inequalities in income, wealth, or opportunites for a higher quality of life are particularly troubled by health inequalities. They believe that a socioeconomic inequality that otherwise seems just becomes unjust if it contributes to health inequalities. Is every health inequality that results from unequally distributed social goods unjust? If there is an irreducible health gradient across socioeconomic groups, does that make the very existence of those inequalities unjust? Does a concern about health inequalities drive us to even more radical forms of social and economic egalitarianism than we had imagined we were committed to?

Alternatively, are some health inequalities the result of acceptable trade-offs? Perhaps they are simply unfortunate by-products of inequalities that work in other ways to help worse-off groups. For example, it is often claimed that permitting inequality provides incentives to work harder, thereby stimulating growth that will ultimately benefit the poorest groups. To whom must these trade-offs be acceptable if we are to consider them just? Are they acceptable only if they are part of a strategy aimed at making the situation more just? In our judgments about justice, does it matter exactly how social determinants produce inequalities in health status?

Unfortunately, these questions have been almost totally ignored within the field of bioethics, as well as within ethics and political philosophy more generally.[4] With some significant exceptions, bioethics has not looked "upstream" from the point of delivery of medical services to the role of the health-care system in improving population health. Bioethics has even more rarely looked further upstream to the distribution of social goods that determine the health of societies (cf. Marchand et al. 1998).

This omission is quite striking, for bioethics here lags behind both social science and policy concerns. The social science literature on social determinants of health has grown impressively for a quarter of a century. More

[4] Sen (2004) is an important exception.

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important, a concern about health equity and its social determinants has emerged as an important consideration in the policies of several European countries over the past two decades (Benzeval et al. 1995; Ostlin and Diderichsen 2001). WHO has devoted growing attention to inequalities in health status and the policies that cause or mitigate them. So have research initiatives, such as the Global Health Equity Initiative, funded by the Swedish International Development Agency and the Rockefeller Foundation (cf. Evans et al. 2001). In the past decade, there has emerged a growing social science and policy concern about health disparities in the United States, although its focus has largely been on issues of race, not class, with a few important exceptions.

To fill this bioethical gap, I shall argue that we must supplement a widely cited intuitive analysis of what health inequalities are unjust – those that are avoidable, unnecessary, and unfair (Dahlgren and Whitehead 1991; Whitehead 1992) – with a more systematic account. In the previous chapter, I turned to Rawls's theory of justice as fairness, as well as to other recent work in distributive justice, to show that there is broad support for obligations to protect opportunity. I drew on that support to back my claim that there is a special moral importance to meeting health needs. Here I turn again to Rawls's theory to illustrate what an account of the just distribution of the determinants of health might look like. My contention is that – quite unintentionally – Rawls's theory provides a defensible account of how to distribute the social determinants of health fairly and thus tells us something useful about when health inequalities are unjust. Specifically, each of his principles of justice as fairness governs a cluster of key social determinants of health. General conformance with these principles will consequently flatten the socioeconomic gradient of health as much as we can reasonably demand (and, arguably, considerably more than we observe even in wealthy nations with developed social welfare protections). *In effect, social justice in general is good for population health and its fair distribution.* There will remain a problem of residual health inequalities produced by otherwise justifiable inequalities, and we shall consider the extent to which justice as fairness gives any guidance about them.

In turning to Rawls to illustrate what an account of the distribution of the determinants of health might look like, my intention, as in the previous chapter, is neither to argue in favor of Rawls's theory nor to presuppose its truth.[5] Nevertheless, the extension of the power and scope of his theory that results from the argument in both chapters is of some methodological interest, a point I return to later. First, however, I shall describe some of

[5] Sen (2004) provides another illustration, arguing that health inequity is clearly connected with social injustice more broadly and is not to be confused with health inequality.

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the findings from recent social epidemiological work and point to their serendipitous connection to Rawls's principles of justice.

SOCIAL DETERMINANTS OF HEALTH: SOME BASIC FINDINGS

Four central findings in the literature on the social determinants of health have implications for an account of justice and health inequalities. First, if we look across nations, the national income/health gradients we observe are not the result of some fixed or determinate laws of economic development but are influenced by social policy choices. The wealth of nations matters up to a relatively modest point, but policies always matter, and arguably they matter more in poor countries. Second, if we look within societies, the individual SES/health gradients we observe are not just the result of the deprivation of the poorest groups. Rather, a gradient in health operates across the whole socioeconomic spectrum within societies even where universal medical coverage and other welfare programs are in place. Third, although there is broad agreement that the degree of social inequality contributes to the steepness of a society's health gradient, there is conflicting evidence and a complex debate about whether the steepness of the gradient is affected by the degree of income inequality in a society. Fourth, there are reasonable hypotheses identifying some social and psychosocial pathways through which inequality affects health. These causal pathways can be molded or changed by specific policy choices that should be guided by considerations of justice, and they point to institutions that are appropriately governed by Rawls's principle of justice as fairness.

Cross-National Evidence on Health Inequalities

The finding that prosperity is related to health, whether measured at the level of nations or individuals, might lead one to the conclusion that these income/health gradients are inevitable. But evidence suggests otherwise. Figure 3.1 shows the relationship between the wealth and health of nations as measured by per capita gross domestic product (GDPpc) and life expectancy. There is a clear association between GDPpc and life expectancy, but only up to a point. The relationship levels off beyond about $8,000 GDPpc, with virtually no further gains in life expectancy. This leveling effect is most apparent among the advanced industrial economies that largely account for the upper tail of the curve in Figure 3.1.

Closer inspection of this figure points out some startling discrepancies. Though Cuba and Iraq (before the second Iraq war) are equally poor (GDPpcs about $3,100), life expectancy in Cuba exceeds that in Iraq by 17.2 years in 1995 (and no doubt by more now). And while the difference between the GDPpc for Costa Rica and the United States is enormous

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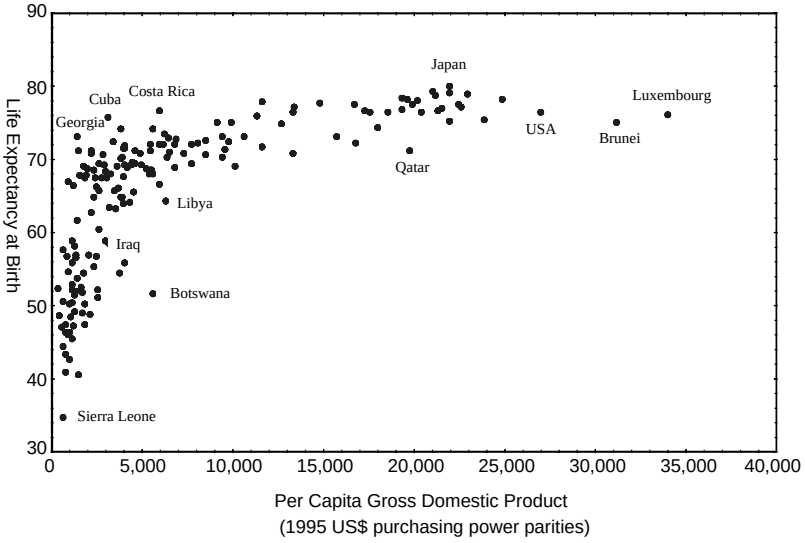


FIGURE 3.1. Relationship between Country Wealth and Life Expectancy. *Source:* United Nations Human Development Report Statistics 1998.

(about $21,000), Costa Rica's life expectancy is nearly the same as that of the United States. In fact, despite being the richest nation on earth, the United States performs rather poorly on health indicators (i.e., forty-fourth in life expectancy at birth)[6].

Taken together, these observations support the notions that the relationship between economic development and health is not fixed and that national health achievement is mediated by factors other than aggregate wealth. Culture, social organization and government policies, among others, are important in determining population health and thus in explaining the differences in health outcomes among nations. Variations in these socially controllable factors are thus important from the perspective of social justice.

Individual SES and Health

At the individual level, numerous studies have documented the "socioeconomic gradient." On this gradient, each increment up the socioeconomic hierarchy is associated with improved health outcomes over the rung below (Black et al. 1988; Davey-Smith et al. 1990; Pappas et al. 1993; Adler et al. 1994). Note that this relationship is not simply a contrast between the health

[6] CIA Factbook, https://www.cia.gov/cia/publications/factbook/rankorder/2102rank.html, accessed July 18, 2006.

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of the rich and the poor, but is observed across all levels of SES (remember that not all the components of SES – income, education, social class – behave identically; see footnote 2). Though the effects of income or wealth work across the SES spectrum, the gradient is steeper at lower income levels, with considerable flattening out at the highest income levels. This "concavity" of the gradient means that transfers of resources from the best-off to the worst-off SES groups would improve aggregate health and would have little negative effect, if any, on the best-off groups. (I return to this point in my later discussion of Rawls's difference principle.)

What is particularly notable about the SES gradient is that it does not appear to be explained by differences in access to health care. Steep gradients have been observed even among groups of individuals, such as British civil servants, with adequate access to health care, housing, and transport (Davey-Smith et al. 1990; Marmot et al. 1998). The Whitehall studies of British civil servants provide strong support for such a gradient, not only for mortality rates and life expectancy, but also for morbidity rates across a range of diseases. Strikingly, some of these studies show that civil service rank explains more of the risk of premature heart disease among ranks than such standard risk factors as smoking, serum cholesterol level, or blood pressure (Marmot et al. 1984).

Importantly, the steepness of the gradient varies substantially across societies. Some societies show a relatively shallow gradient in mortality across SES groups. Others, with comparable or even higher levels of economic development, show steeper gradients in mortality rates across the socioeconomic hierarchy.

Relative Income and Health

A lively debate exists in the empirical literature about whether income inequality, as opposed to absolute levels of income, helps to determine the steepness of the gradient. Were this true, we might find that middle income groups in a country with high income inequality have lower health status than comparable or even poorer groups in a country with low income inequality. There is some evidence for this pattern in the United States if variations among states in inequality are examined (Kennedy et al. 1998; Lynch et al. 1998; Subramanian and Kawachi 2006), but there are conflicting study results as well (Mellor and Milyo 2002; Deaton and Lubotsky 2003). This effect, if it can be established, is apparent in Figure 3.2, where the prevalence of self-reported fair/poor health is higher for almost every income group (and the gradient is steeper) for those living in states with the highest income inequality (Kennedy et al. 1998). This effect of shifting the curve, if it is supported by the empirical findings, would suggest that income inequality has an effect over and above what is implied by the concavity of the SES gradient itself.

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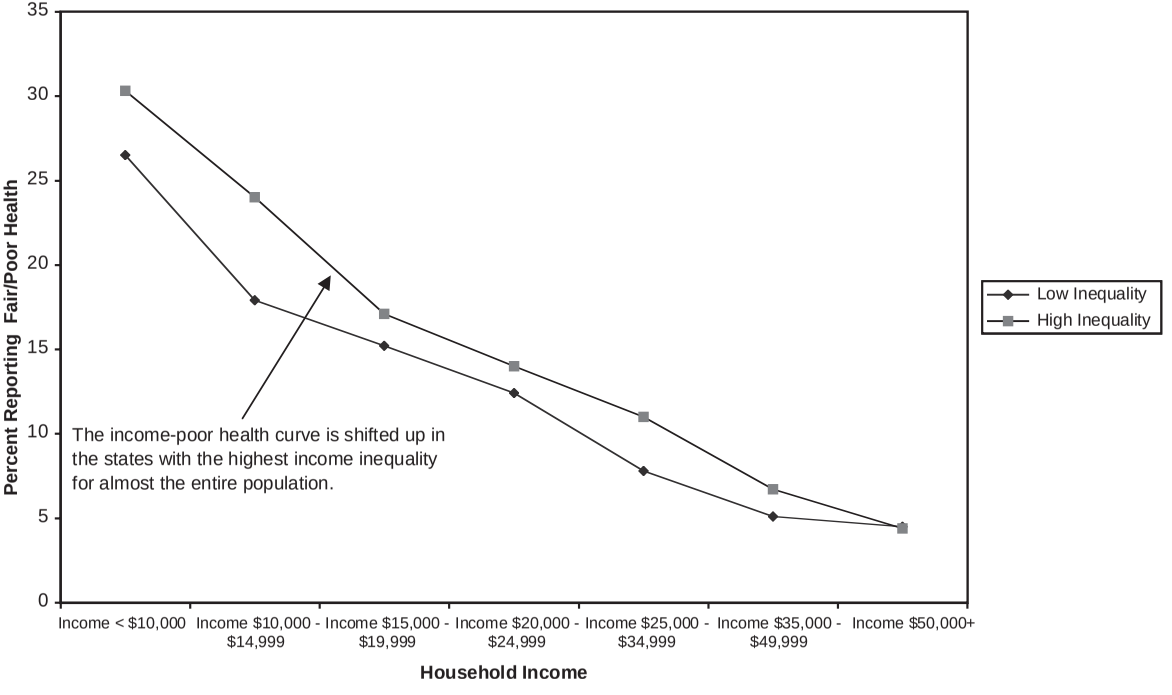


FIGURE 3.2. Self-Rated Health and Individual Household Income.

Leaving aside the contentious results of these studies,[7] it is worth noting why the income relativity thesis has attracted such interest. If it is true, it is not just the size of the economic pie but how the pie is shared that matters for population health. It is not the absolute deprivation associated with low economic development (lack of the basic conditions necessary for health, such as clean water, adequate nutrition and housing, and general sanitary living conditions) that explains health differences among developed nations but the degree of *relative deprivation* within them. Relative deprivation refers not to a lack of the goods that are basic to survival, but rather to a lack of sources of self-respect that are deemed essential for full participation in society. (These "recognitional" features of just arrangements are emphasized in Rawls's account, as we shall see.) If the income relativity thesis can be established, it would then require elaboration of the causal mechanisms that underlie it.[8]

The income relativity thesis is interesting and worth further investigation, but it has only a modest bearing on the claims I pursue in this chapter. As we shall see, there is strong evidence that health inequalities are reduced if we concentrate first on making the worst-off groups socially and economically as well off as possible, paying special attention to improvements in human capital and investments in public health. If true, the income relativity thesis

[7] Sparked by Wilkinson's (1992) paper showing a correlation between national measures of income inequality and the gradient of health inequality among a select groups of Organization of Economic Cooperation and Development (OECD) countries, researchers in the United States and elsewhere found conflicting results during the subsequent decade. The initial positive results from the OECD countries disappear when a full rather than a selected set of them is studied. Negative results are also found primarily in countries more egalitarian than the United States (Sweden, Japan, Canada, Denmark, New Zealand), as noted by Subramanian et al. (2003) and Subramanian and Kawachi (2004). Some negative results are also found in the United States, especially if units of aggregation are smaller, such as municipalities rather than states. A significant number of U.S. studies support the claim, especially at the state level. Positive results are also found in a recent study in Chile, a country with more income inequality than exists in the United States (Subranamian et al. 2003).

This pattern of seemingly conflicting results might be explained if income inequality affects health inequalities only above some threshold level of inequality, a threshold crossed in the United States and more unequal countries but not in more egalitarian countries (Subramanian and Kawachi 2004). Alternatively, the effect of income inequality might be countered by more egalitarian social policies, such as the more redistributive health and welfare policies in other OECD countries. This latter alternative highlights an important point about the claim generally made by supporters of the income relativity thesis: Income inequality is a causal factor that works through various mechanisms, political, social, and possibly individual (e.g., stress), and is not claimed to be "by itself" or "per se" a cause of health inequalities.

[8] In the United States, in those studies that support the thesis, the strength of this relationship between social inequality and health inequality is not trivial. One study of U.S. metropolitan areas, rather than states, found that areas with high income inequality had an excess of death compared to areas of low inequality that was equivalent in magnitude to all deaths due to heart disease (Lynch et al. 1998).

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would strengthen support for giving priority to making the worst off as well off as possible, for it would make an even stronger claim that it works to the advantage of all. In addition, it might point us to other mechanisms that we may have to address.

Pathways Linking Social Inequalities to Health Inequalities

To address key issues about social policy and distributive justice, we must better understand the mechanisms underlying the correlations between health outcomes and social inequalities in income, education, and other factors. Unless we know something about these underlying causal mechanisms, we cannot focus properly on the socially controllable factors that a just policy must regulate. Fortunately, there are plausible and researchable pathways through which social inequalities produce inequalities in health.

The first empirical result I highlighted earlier was the claim that social policy, more than the wealth of nations, explains variations in health outcomes across countries. Developing countries that invest heavily in human capital, for example in education, have better health outcomes. Indeed, one of the strongest predictors of life expectancy among developing countries is adult literacy, particularly the disparity between male and female adult literacy, which explains much of the variation in health achievement among these countries after accounting for GDPpc. Similarly, in the United States, differences between the states in women's status – measured in terms of their economic autonomy and political participation – are strongly correlated with with female mortality rates (Kawachi and Kennedy 1997; Kawachi et al. 1999). Even if we do not know all the underlying mechanisms through which improved education, including female literacy, improves health outcomes, we have good evidence that investment in this socially controllable factor has measurable health effects.

The second empirical thesis I highlighted concerned the strong effect of a socioeconomic gradient of health. This gradient is demonstrated even where socioeconomic inequality does not mean absolute deprivation in material goods or access to health care – as in the Whitehall studies. Indeed, if we focus on the British civil servants, one important variable in their job situations concerns their degree of control over work (Marmot 2004). Such control may affect the kinds of stress people are exposed to, and this in turn may affect basic immune mechanisms or may directly impact the course of certain diseases. Uncovering mechanisms here is a complex research program that may require work on both animals and humans, including some carefully selected natural experiments. How readily we can convert this understanding of mechanisms into modifications of social policy, including, for example, the organization of the workplace, remains to be seen.

Our third thesis concerned the controversial evidence about the degree of income inequality and its effects on health. It may be premature to look

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for mechanisms since the basic association is controversial. Still, we may better understand some of the mechanisms through which income inequality might work if we note some of the related studies.

In the United States, the states with the most unequal income distribution invest less in public education, have more uninsured persons, and spend less on social safety nets (Kaplan et al. 1996; Kawachi and Kennedy 1997). Differences in human capital investment are particularly striking. These are demonstrated for educational spending and, more importantly, for outcomes; even when controlling for median income, income inequality explains about 40 percent of the between-state variation in the percentage of fourth-grade children who are below the basic reading level. Similarly strong associations are seen for high school dropout rates. These data clearly show that educational opportunities for children in states with high income inequality are quite different from those in more egalitarian states. Furthermore, these early effects not only have an immediate impact on health, increasing the likelihood of premature death during childhood and adolescence (as evidenced by the much higher death rates for infants and children in the high-inequality states), but also have lasting effects showing up later in life as part of the SES gradient in health (Bartley et al. 1997; Davey-Smith et al. 1990).

These societal mechanisms are tightly linked to the political processes that influence government policy. For one thing, income inequality appears to affect health by undermining civil society. Income inequality erodes social cohesion, increasing social mistrust and reducing participation in civic organizations – two features of civil society (Kawachi and Kennedy 1997; Kawachi et al. 1997). Lack of social cohesion, in turn, is reflected in significantly less political activity (e.g., voting, serving in local government, volunteering for political campaigns), thus undermining the responsiveness of government institutions in addressing the needs of the worst-off groups. This is demonstrated not only by the human capital investment data presented earlier but also by the lack of investment in human security. States with the highest income inequality, and thus the lowest levels of social capital and political participation, provide far less generous social safety nets.

AN INTUITIVE ANALYSIS OF HEALTH INEQUITIES

In the public health literature (Braveman 1999), there is an influential intuitive answer to our second Focal Question, namely, Whitehead's (1992) and Dahlgren and Whitehead's (1991) claim that health inequalities count as inequities when they are *avoidable*, *unnecessary*, and *unfair*. How do these conditions fit together?

Presumably, inequalities that are avoidable and unnecessary but not unfair do not count as inequitable. For example, if lung cancer rates differed between smokers and nonsmokers, and if smoking rates were not themselves

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strongly influenced by SES, ethnicity, or gender but seemed completely a matter of individual, voluntary, informed choice, then we might not think the cancer rate inequality unfair and thus not inequitable. Similarly, an inequality may be unfair but also unavoidable or even necessary, given our inability to change certain conditions. For example, suppose we learned that the mortality rate of certain workers, say truck drivers, means that their life expectancy is lower than that of other workers, and many believe (intuitively) that this is unfair. We may be able to reduce that higher mortality rate through road and equipment improvements and through better training of drivers and enforcement of traffic laws, but suppose there remains an elevated mortality rate that seems unavoidable, or perhaps necessary given the importance of trucking. Some may insist that the inequality is still unfair, if unavoidable, even though they would conclude that the inequality is not an inequity. If we can agree on what is avoidable, unnecessary, and unfair, and if this analysis is correct, then we can agree on which inequalities are inequitable.

The Whitehead–Dahlgren analysis is deliberately broad. It not only includes socioeconomic differences but others as well. Age, gender, race, and ethnic differences in health status exist that are independent of socioeconomic differences, and they raise distinct questions about equity or justice. For example, should we view the lower life expectancy of men compared to women in developed countries – as much as six years' difference in some observations – as an inequity?[9] If it is rooted in biological differences that we do not know how to overcome, then, according to this analysis, it is not avoidable and therefore not an inequity. Perhaps not all of it is unavoidable – higher smoking rates for men or higher workplace risks may explain some of the difference. But how much? This is not an idle controversy: Taking average rather than gender-differentiated life expectancy in developed countries as a benchmark or goal will yield different estimates of the inequity women face in some developing countries.[10] In any case, the analysis of inequity here is only as good as our understanding of what is avoidable or unnecessary.

The same point applies to judgments about fairness. Is the poorer health status of a social class or ethnic group that engages in heavy drug and alcohol use unfair? We may be inclined to say it is not unfair, provided that the use or avoidance of drugs and alcohol is truly voluntary. But if many people

[9] Sen (2004) points to the gender inequality and denies that it is an inequity, not because we could not do something about it but because what we could do, e.g., give men more health care than women, would be discriminatory, violating "process" constraints on fair treatment.

[10] Murray's reduction of the male–female gender gap to 2.5 years from an observed 6 years, based on extrapolation from some cases and assumptions about greater exposure to risk, was criticized as possible gender bias in the construction of DALYs. See Anand and Hanson (1998).

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in an ethnic group or class behave similarly, there may also be factors at work that reduce the autonomous nature of their behavior and how much responsibility they should bear (Wikler 1978; Roemer 1995). The analysis thus leaves us with judgments about responsibility that are complex and unresolved; as a result, we have continuing disagreements about fairness (or avoidability).

The poor in many countries have no clean water, sanitation, adequate shelter, basic education, vaccinations, and prenatal and maternal care. As a result of some or all of these factors, there are infant mortality differences between them and richer groups. Since social policies could supply the missing determinants of infant health, the inequalities are avoidable.

Are these inequalities also unfair? Most of us would immediately think that they are, perhaps because we believe that international (Pogge 2002) or domestic policies that create and sustain poverty are unjust. We also believe that social policies that compound poverty with lack of access to the determinants of health are doubly unfair. Of course, libertarians would disagree. They would insist that what is merely unfortunate is not unfair; on their view, we have no obligation of justice, as opposed to charity, to provide the poor with what they are missing. Many of us might reject the libertarian view as itself unjust because of this dramatic conflict with our beliefs about poverty and our social obligations to meet people's basic needs.

The problem becomes more complicated, however, when we remember one of the basic findings from the literature on social determinants: We cannot eliminate health inequalities simply by eliminating poverty. Health inequalities persist even in societies where poor people have access to all the determinants of health noted previously. Furthermore, these health inequities persist as a gradient of health throughout all SES classes, not just between the very poorest groups and those immediately above them. Some of the mechanisms may be difficult to avoid and may depend on deep structural features of an economy. Thus, educational and skill differences that steer people into hard manual labor rather than nonmanual labor significantly affect the rate at which poor health is self-reported (Case and Deaton 2003). Whether we consider these mechanisms and their effects unfair may depend in part on our views about the fairness of overall social inequality.

At this point, many of us are forced to reexamine what we believe about the justice of the remaining socioeconomic inequalities. Unless we believe that *all* socioeconomic inequalities (or at least all inequalities we did not choose) are unjust – and very few do so – we must consider more carefully the problem created by the health gradient and the fact that it becomes steeper as inequality increases. Our judgments about the intuitive notion of fairness in the Dahlgren–Whitehead account provide less guidance in thinking about the broader issue of the social determinants of health inequalities even if we, rightly or wrongly, felt confident in appealing to them when rejecting the libertarian position. Indeed, we may even believe that some

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socioeconomic inequality is unavoidable or even necessary and therefore is not unjust.

A RAWLSIAN ANALYSIS OF HEALTH INEQUITIES

One reason we develop general ethical theories, including theories of justice, is to provide a framework for settling important disputes about conflicting moral beliefs or intuitions of the sort facing the Whitehead–Dahlgren analysis. For example, in *A Theory of Justice*, Rawls sought to leverage our relatively broad liberal agreement on principles guaranteeing certain equal basic liberties into an agreement on a principle limiting socioeconomic inequalities, a matter on which liberals considerably disagree (J. Cohen 1989). His strategy was to show that a social contract designed to be fair to free and equal people ("justice as [procedural] fairness") would not only justify the choice of those equal basic liberties but would also justify the choice of principles guaranteeing equal opportunity and limiting inequalities to those that work to make the worst-off groups fare as well as possible.

Rawls's account, though developed to answer this general question about social justice, also provides principles for the just distribution of the social determinants of health, unexpectedly adding to its scope and power as a theory. The extra power of the theory is a surprise, since, as I emphasized earlier, Rawls deliberately simplified the construction of his theory by assuming that his contractors are fully functional over a normal life span. Examining the social determinants of health inequalities from the perspective of Rawls's theory is particularly appealing because justice as fairness is egalitarian in orientation (Daniels 2003) and yet justifies certain inequalities, such as those in income and wealth, that contribute to health inequalities. In addition, my extension of Rawls links the protection of health to the protection of equality of opportunity, again setting up the potential for internal conflict. To see whether this combination of features leads to contradictions in the theory or to insight into the problem, we must examine the issue in more detail.

Rawls on Allowable Inequalities

How does Rawls justify socioeconomic inequalities? Why wouldn't free and equal contractors simply insist on strictly egalitarian distributions of all social goods, just as they insist on equal basic liberties and equal opportunity?

Rawls's answer is that it is *irrational* for contractors to insist on equality if doing so would worsen their lifetime prospects.[11] If incentives to develop

[11] This reasoning from the perspective of Rawlsian contractors is the central argument in Rawls and is to be contrasted with the "intuitive" argument discussed in Chapter 2 that turned on the moral arbitrariness of the natural distribution of talents and skills.

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skills and take risks increase social productivity, then the social pie that gets divided may be increased so much that less than equal shares for those who are worst off are still absolutely larger than equal shares of a smaller social pie. Specifically, Rawls argues that contractors would choose the difference principle, which, as we noted in Chapter 2, permits inequalities provided that they make the worst-off groups in society as well off as possible.[12] The argument for the difference principle appears to suggest that relative inequality is less important than absolute well-being, a suggestion that is in tension with other aspects of Rawls's view. The tension is illustrated by Rawls's insistence that inequalities allowed by the difference principle should not undermine the value of political liberty or the requirements of fair equality of opportunity. The priority given these other principles over the difference principle thus limits the inference that Rawls has no concern about relative inequality. Specifically, as we shall see, these principles work together to constrain inequality and to preserve the social bases of self-respect for all.

Two points will help avoid misunderstanding of the difference principle and its justification. First, as we saw in Chapter 2, it is not a mere trickle-down principle, but one that requires maximal flow to help the worst-off groups. The worst off, and then the next worst off, and so on (Rawls [1971] calls this "chain connectedness") must be made as well off as possible, not merely somewhat better off, as a trickle-down principle implies. The difference principle is thus much more demanding than a principle that would permit any degree of inequality provided that there was some trickle of benefits to the worst off.[13] Indeed, it is more egalitarian than alternative principles that merely assure the worst off a "decent" or "adequate" minimum. Part of the rationale for the more demanding principle is that it would produce less strain of commitment, less sense of being unfairly left out, at least for those who are worst off, than principles that allow more inequality (J. Cohen 1989). Indeed, from what we have learned about the social determinants of health, the more demanding difference principle would also produce less

[12] A careful discussion of Rawls's argument for the difference principle and the extensive critical literature it has generated is beyond the scope of this chapter. It is important, however, to distinguish Rawls's own social contract argument from the many informal and intuitive reformulations of it. See Barry (1989) and G. A. Cohen (1992, 1995). Some controversy is fed by Rawls's (1971: Sect. 13) different formulations of the difference principle; some comes from the complexity of other assumptions involved in his argument, such as "chain connection" and "close-knittedness"; some follows from disagreement with giving such strong priority to the worst off, regardless of the benefits lost to others and regardless of how well off the worst off are. For a useful discussion, see Williams (1995).

[13] What was said previously applies to the ideal case, where a society is in general conformance with the principles of justice. In a nonideal setting, where we may be making a system more just, Rawls is willing to say that an improvement is not unjust as long as it moves the worse-off groups closer – even if not maximally closer – to being as well off as possible.

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health inequality than any other principles that allow inequalities. By flattening the health gradient, it also benefits middle-income groups, not simply the poorest. In this regard, its benefits are important beyond the level where we have helped the worst off to achieve "sufficiency." This point provides a reply to those who suggest that the difference principle has no appeal once the worst off are sufficiently provided for (Gutmann and Thompson 1996).

Second, when Rawlsian contractors evaluate how well off the principles they choose will make them, they are to judge their well-being by an index of "primary social goods" (see Chapter 2; also see Rawls 1971: 62; 1993: Ch. 5). The primary social goods, which Rawls thinks of as the "needs of citizens," include liberty, powers, opportunities, income and wealth, and the social bases of self-respect. (These objective measures of well-being should be contrasted with measures of happiness or desire satisfaction that are familiar from utilitarian and welfare economic perspectives.) In his exposition of the difference principle, Rawls illustrates how it will work by asking us to consider only the simpler case of income inequality. In doing so, he *assumes* that the level of income will correlate with the level of other social goods on the index.

This simplification can be misleading. In crucial cases – including those involving health – the correlation may not occur and it will be essential to use the whole index, rather than just the income component, to measure how well off people are.[14] For example, let us suppose that having "democratic" control over one's workplace is crucial to self-realization and the promotion of self-esteem (J. Cohen 2001). Marmot (2004) documents how hierarchical workplaces may create health inequalities through the effect of status on stress. Suppose further that hierarchical workplaces are more efficient than democratic ones, providing higher incomes for the worst-off workers than democratic workplaces. Then the difference principle does not clearly tell us whether the hierarchical workplace contains allowable inequalities since the worst off are better off in some ways (economic) but worse off in others (health and thus opportunity and self-respect). Without knowing the weighting of items in the index, we cannot use it to say clearly what inequalities are permitted. When we evaluate which income inequalities are allowable, by asking which ones work to make the worst-off groups as well off as possible, we must judge how well off groups are by reference to the *whole* index of primary goods and not simply the resulting income.

This point is of particular importance in the current discussion. My extension of Rawls in Chapter 2 treats health status as a determinant of the opportunity range. Since opportunity is included in the index of primary social goods, the effects of health inequalities are thereby included as well. I also

[14] In thinking about the impact of income inequality on health, Deaton (personal communication, July 2006) makes a related point when he cautions that we should think more broadly about social inequality and not income alone.

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noted earlier that the social bases of self-respect are among the primary social goods on the index. From Marmot's (2004) work, we may infer that hierarchical workplaces not only undermine health, and thus opportunity, but do so because there is inadequate protection of self-respect. Clearly, we must use the whole index, not the simplifying assumption that income correlates with all other items on it.

Unfortunately, Rawls says very little about how items in the index are to be weighted. This is one of the crucial points on which the theory says less than we might have wished. Therefore, we have little guidance about how these primary goods are to be traded off against each other in constructing the index. This silence pertains not only to the use of the index in the contract situation, but also to its use by a legislature trying to apply the principles of justice in a context where many features of a society are known.

Flattening the SES Gradient of Health

Now that we have seen why Rawls believes that inequalities are allowable under some conditions, we can consider how those general inequalities affect the distribution of health and contribute to health inequalities. If allowable general inequalities magnify health inequalities, we might conclude that justice, as Rawls conceives it, is not good for our health. Instead, we shall see why social justice, as described by Rawls's principles, is good for our health and promotes its fair distribution.[15] This will permit us to give a more satisfying answer to the second Focal Question.

To understand this claim, let us start with the ideal case, a society governed by Rawls's principles of justice as fairness that seeks to achieve what Rawls calls "democratic equality" (Rawls 1971; Daniels 2003). Consider what the society requires in distributing the social determinants of health. In such a society, all are guaranteed equal basic liberties (Rawls 1982a), including effective exercise of (or worth of) political participation (Daniels 1975; cf. Rawls 1995). In addition, there are institutional safeguards aimed at assuring all, richer and poorer alike, the value of effectively exercisable political participation rights. Without such protection, basic capabilities of citizens cannot develop. Recognizing that all citizens have these capabilities protected is critical to preserving self-esteem, on Rawls's view. In requiring institutional support for political participation rights, Rawls (1995) rejects the claim that freedom of speech of the rich is unfairly restricted by limiting their personal expenditures on their own political campaigns, a limitation the Supreme Court ruled unconstitutional in *Buckley* v. *Valeo*. After all, the limitation does not unduly burden the rich compared to others. There is evidence that political participation is itself a social determinant of health.

[15] Sen (2004) too makes the point that social justice broadly understood is crucial to identifying and addressing health inequity.

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Therefore, the Rawlsian ideal assures institutional protections that counter the usual effects of socioeconomic inequalities on participation and thus on health.

The Rawlsian ideal of democratic equality also involves conformity with a principle guaranteeing fair equality of opportunity (see Chapter 2 for a discussion of this principle). Not only is discrimination prohibited by the principle, but it requires strong measures to mitigate the effects of socioeconomic inequalities and other social contingencies on opportunity. In addition to equitable public education, such measures would include developmentally appropriate day care and early childhood interventions to promote the development of capabilities independently of the advantages of family background. Such measures match or go beyond the best models of such interventions we see in European efforts to provide day care and early childhood education. We also note that the strategic importance of education for protecting equal opportunity has implications for all levels of education, including access to graduate and professional education and continuing adult education in a world where work skills must be continuously upgraded.

The equal opportunity principle also requires extensive public health, medical, and social support services aimed at promoting normal functioning for all, as we saw in Chapter 2. It even provides a rationale for the social costs of reasonable accommodation of persons with incurable disabilities, as required by the Americans with Disabilities Act (see Chapter 5). Because the principle aims at promoting normal functioning for *all* as a way of protecting opportunity for all, it aims at both improving population health and reducing health inequalities. Obviously, this requires universal comprehensive health care, including public health, primary health care, and medical and social support services.

To act justly in health policy, we must understand how socioeconomic (and other) inequalities work to produce differential health outcomes. Suppose that, as Marmot (2004) argues, structural and organizational features of the workplace that induce stress and loss of control tend to promote health inequalities. If this is true, then those features should be modified to reduce their negative effects on health as a public health requirement of the equal opportunity approach; this is on a par with the requirement to reduce exposure to toxins in the workplace (see Chapter 7). Moreover, modifying such features would be favored by Rawls's emphasis on securing the social bases of self-respect.

Finally, in the ideal Rawlsian society, the Difference Principle significantly reduces allowable inequalities in income and wealth and, more generally, in the index of primary social goods.[16] The inequalities allowed by this

[16] G. A. Cohen (1992) has argued that a *strict* interpretation of the difference principle would allow few incentive-based inequalities; for a more permissive view, see Daniels (2003).

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principle (together with the principles assuring equal opportunity and the value of political participation) are probably more constrained than those found in even the most industrialized societies. If so, then the inequalities that conform to the difference principle would produce a flatter gradient of health inequality than we currently observe in even the more extensive welfare systems of Northern Europe.

Earlier, I noted the lively ongoing debate about the effects of income inequality on health inequalities. I believe my claim that the difference principle would flatten gradients more than principles calling for weaker constraints on inequality is true even if the income relativity thesis turns out not to be supported by the evidence. The difference principle is about how well off the worst off are in absolute terms, and though applying it may reduce relative inequality, it would affect mainly health inequalities even if the income relativity thesis does not hold. Those who oppose the income relativity thesis in any event endorse the basic facts about a socioeconomic gradient of health. Pulling the bottom up in absolute terms – coupled with other social policies of the sort assured by the other principles of justice as fairness – would therefore flatten the gradients. If income relativity does have a further effect on health inequalities, then my claim is strengthened further.

In short, Rawls's principles of justice as fairness regulate the distribution of the key social determinants of health, including the social bases of self respect. Nothing about the theory, or my extension of it in Chapter 2, should make us focus narrowly on medical services. Properly understood, justice as fairness tells us what justice requires in distributing all social determinants of health. Nor, I should add, does the fact that we should focus on the distribution of goods in addition to health care mean that health care is no longer of special moral importance. Not only does health care make its own significant contribution to population health, but even if all the other socially controllable factors were properly distributed, some pathology would remain and we would have to respond to its impact on opportunity. Health care may not be the only good of special moral importance, given the moral importance of health, but it remains a good of special moral importance.

Residual Inequalities: Unjust or Not?

We still face an interesting theoretical issue, as well as many practical issues in our nonideal or partially just world. First, the theoretical issue. Even if the Rawlsian distribution of the determinants of health flattens health gradients further than those observed in the most egalitarian developed countries, we must still expect some health inequalities. In part, this may happen because we may not understand all the relevant causes or the interventions for modifying them. The theoretical issue is whether the theory requires us to reduce

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further those otherwise justifiable inequalities because of the inequalities in health status they create.

We should not further reduce those inequalities if doing so reduces productivity so much that we can no longer support the institutional measures we already employ to promote health and reduce health inequality. Our commitment to reducing health inequality should not require steps that threaten to make health worse off for those with less-than-equal health status. So, the theoretical issue reduces to this: Would it ever be reasonable and rational for contractors to accept a trade-off in which some health inequality is allowed in order to produce some nonhealth benefits for those with the worst health prospects?

We know that in real life people routinely trade health risks for other benefits. They do so when they commute longer distances for a better job or take a ski vacation. Some such trades raise questions of fairness. For example, when is hazard pay a benefit workers gain only because their opportunities are unfairly restricted, and when is it an appropriate exercise of their autonomy (see Chapter 7)? Many such trades are ones we think it unjustifiably paternalistic to restrict; others we see as unfair.

Rawlsian contractors, however, cannot base such trades on specific knowledge of their own values. They cannot decide that their enjoyment of skiing makes it worth the risks to their knees or necks. To make the contract fair to all participants, and to achieve impartiality, Rawls imposes a thick "veil of ignorance" that blinds them to all knowledge about themselves, including their views of the good life. Instead, they must judge their well-being by reference to the index of primary social goods (noted earlier) that includes a weighted measure of rights, opportunities, powers, income and wealth, and the social bases of self-respect. But our theoretical question about residual health inequalities reminds us that the theory says too little about the construction of the index to provide us with a clear answer.

One of Rawls's (1971) central arguments for singling out a principle protecting equal basic liberties and giving it (lexical) priority over his other principles of justice is his claim that once people achieve some threshold level of material well-being, they will not trade away the fundamental importance of liberty for other goods. Making such a trade might deny them the liberty to pursue their most cherished ideals, including their religious beliefs, whatever they turn out to be. Can we make the same argument about trading health for other goods?

There is some plausibility to the claim that rational people should not trade health for other goods. Loss of health may prevent us from pursuing what we most value in life. We do, after all, see people willing to trade almost anything to regain health once they lose it.

If we take this argument seriously, we might conclude that Rawls should give opportunity, including the effects of health status, a heavier weighting

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in the construction of the index than income alone.[17] Such a weighting would mean that absolute increases in income for the worst off that might otherwise have justified increasing relative income inequality, according to the difference principle, now fail to justify those inequalities because of the negative effects on opportunity. Although the income of the worst off would increase, they are not better off according to the whole (weighted) index of primary social goods, and so the greater inequality is not permitted. Rawls's simplifying assumption that income correlates with other goods fails in this case (as it did in the hypothetical example of workplace democracy cited earlier).

Nevertheless, there is also strong reason to think that the priority given to health, and thus opportunity, is not as clear-cut as the previous argument implies, especially where the trade is between a *risk* to health and other goods that people highly value. Refusing to allow any (ex ante) trades of health risks for other goods, even when the background conditions are otherwise fair, may seem unjustifiably paternalistic, perhaps in a way that refusal to allow trades of basic liberties is not.

I propose a pragmatic route around this problem, one that has a precedent elsewhere in Rawls. Fair equality of opportunity, Rawls admits, is only approximated even in an ideally just system, because we can only mitigate, not eliminate, the effects of family and other social contingencies (Fishkin 1983). For example, only if we were willing to violate widely respected parental liberties could we intrude in family life and "rescue" children from parental values that arguably interfere with equal opportunity. Similarly, though we give general priority to equal opportunity over the difference principle, we cannot achieve complete equality in health any more than we can achieve completely equal opportunity. Even ideal theory does not produce perfect justice. Justice is always rough around the edges. Specifically, if we had good reason to think that "democratic equality" had flattened inequalities in accord with the principles of justice, then we might think we had done as much as was reasonable to make health inequalities fair to all. The residual inequalities that emerge with conformance to the principles are not a compromise with what justice ideally requires; they are acceptable as just.

So far, we have been considering whether the decision about such a trade-off can be resolved from the perspective of individual contractors. Instead, suppose that the decision is to be made by the legislature in a society that conforms to Rawls's principles. Because those principles require effective political participation by all socioeconomic groups, we can suppose that

[17] Rawls (1971: 93) does suggest that since fair equality of opportunity is given priority over the difference principle, within the index we can assume that opportunity has a heavier weighting.

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the groups most directly affected by any trade-off decision have a voice in the decision. Since there is a residual health *gradient*, groups affected by the trade-off include not only the worst off, but those in the middle as well. Developing a democratic process that involved deliberation about the trade-off and its effects might be the best we could do to answer the theoretical question (see Chapter 4).

In contrast, where the fair value of political participation is not adequately assured – and we doubt that it is so assured in even the most democratic societies – we have much less confidence in the fairness of a democratic decision about how to trade health against other goods. It is much more likely that those who benefit most from the inequalities, that is, those who are better off, also wield disproportionate political power and will influence decisions about trade-offs to serve their interests. It may be that the use of a democratic process in nonideal conditions is the fairest resolution we can practically achieve, but it still falls well short of what an ideally just democratic process would involve.

A BRIEF REMARK ON METHOD

The examination of practical questions in ethics has implications for ethical theory, since influence is not just unidirectional. Rawls's "ideal" theory was formulated in a way that assumed away problems of health. By examining what justice requires in health care, it was possible to revise the theory so that the idealizing assumption could be dropped, thus extending its power. In turn, the extended theory informs practical deliberation about institutional design (see Chapter 9). By connecting recent empirical work on social determinants with the normative theory, we further extend its power.

There may be more here than an extension of scope. When an empirical theory turns out to explain new phenomena that were not part of the evidentiary base for its laws, we tend to conclude that the concepts incorporated in it are "projectible" in a desired way. We may think of them as better confirmed as a way of dividing up or describing the world. Rawls began with certain political concepts thought crucial to our well-being – to meeting our needs as free and equal citizens of a democracy. Using these ideas, the goal was to select terms of cooperation that all free and equal citizens could agree are fair and reasonable. It then turns out, in light of the social science literature, that the aspects of well-being captured by these ideas expand to include the health of the population as well. Whatever controversy may surround some of these political components of well-being, they do connect – albeit empirically – to an incontrovertibly objective component of our social well-being, namely, the health of the population. If this were an empirical rather than a normative theory, we would think that the evidence of projectibility counted in its favor and constituted support for the theory. Is there additional support for Rawls's theory?

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